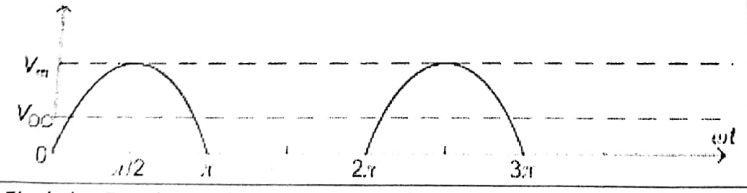
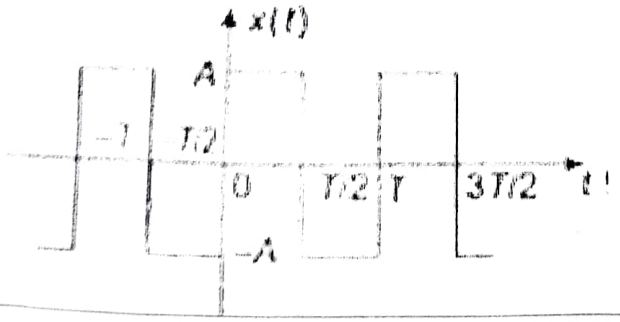
(Please write your Enrollment Number) Enrollment No. 02701182025

MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: 1st)
(October, 2025) OFF LINE mode

Subject Code: BEC 102	**Subject: Signals and Systems**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

		(2.5*4=10)	CO Mapping
Q1			
	a) What are the key Differences between a continuous-time Signal and a discrete-time signal		**CO1**
	b) Explain Parseval's Theorem with proof.		**CO1**
	c) State the relationship between the step response and the impulse response of an LTI system.		**CO2**
	d) Draw and explain the ideal characteristics of a Low-Pass Filter (LPF), High-Pass Filter (HPF), and Band-Pass Filter (BPF).		**CO2**
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a)Sketch the waveforms-: i. x(t)= u(t+1)-2u(t-1)+2u(t-2) ii. x(t)= r(t+1)-r(t)+r(t-2)		**CO1**
	b) Check whether the System is -: i. y(t)=x(logt) (Linear/NonLinear) ii. y(t)=x(4t) (Time-Variant/Time-Invariant)		**CO1**
	c) Explain the terms with Proof-: i. BIBO stability ii. Energy Density Spectrum		**CO1**
	UNIT II		**CO Mapping**
Q3	***Attempt any two parts***	**(5*2=10)**	
	a) *Determine the complete response of the system describes with the given differential equation, with initial conditions:* $y(0) = 1$, $dy(t)/dt\|_{(t=0)} = -1$, *for the input:* $x(t) = e^{(-t)}\,u(t)$: $d^2y(t)/dt^2 + 5\,dy(t)/dt + 6y(t) = 3x(t)$		**CO2**
	b) *Find the Fourier series expansion of half wave rectified sine wave.* 		**CO2**
	c) *Find the Fourier series expansion of the given waveform for T=* 2π: 		**CO2**